

Genetic load in
Drosophila melanogaster

- Work in progress -

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Abstract

Here, we analysed differences in the impact of deleterious variants on fitness depending on the chromosome location of genotypes, using publicly available data resources. It has long been suggested that the X chromosome accumulates male-beneficial genotypes at an accelerated rate, as selection acts more efficiently on the hemizygous sex. In this regard the X chromosome can be seen as an analogy for the haploid males of eusocial insects, as they are hemizygous for every chromosome. If recessive deleterious alleles become exposed, for example during continuous inbreeding, a reasonable assumption is that the deleterious genotypes elicit a less severe decrease in fitness in males than in females. We used the X chromosome as a proxy for the haploid males, and tested this hypothesis by first identifying putatively deleterious variants in the genotypes of heavily inbred *Drosophila melanogaster* flies. Then we calculated the genetic load of said putatively deleterious variants, a measurement of the realised and masked burden caused by deleterious minor alleles. Lastly, we analysed the relationship between fitness and genetic load, taking the chromosome location of the variants and sex and life stage of the flies into account. We found no decrease in fitness with increasing genetic load and little impact of chromosome location and sex. This may be explained by a strong genetic purging effect caused by repeated inbreeding but also by technical limitations in the data structure we have encountered and discussed. However, this project did not fail the analogy between the X chromosome and male haploids, nor did it delegitimise the hypotheses in question. Instead, it provides a valuable basis for future research by providing freely accessible and reproducible methods and highlighting what needs to be paid attention to when choosing the data to address this research topic.

Introduction

The concept of evolution is based on the principle that organisms are subject to variation and mutation, heritability and selection. Whereas the different modes of selection are stochastic processes which pull individuals or populations towards fitness optima, the majority of variation in the form of mutation has been suggested as neutral (Kimura, 1968, 1991) or nearly neutral (Ohta, 1992) to the organism and governed by random genetic drift. Whether through mutation or migration, deleterious alleles constantly invade gene pools. Across multiple contemporary populations, conservative estimations of mutation rates (regardless of effect), μ , are on scales of $0 < \mu \leq 2.5 \times 10^{-8}$ mutations per site and generation (Nachman & Crowell, 2000; Popovic et al., 2024; Scally, 2016), although these values vary greatly depending on location and mutation type. Excluding the most severe alleles which will quickly be selected against and removed, the accumulation of disadvantageous alleles becomes inevitable in genomes with thousands or millions of mutable sites (Agrawal & Whitlock, 2012).

Considering mutation-selection balance, in which mutation constantly introduces deleterious mutations and selection removes them in an equilibrium, Haldane discussed the effects of variation on fitness (Haldane, 1937). He hypothesised that relative to an individual free of mutation, the approximated loss of fitness caused by mutation is $L = 1 - e^{-U}$, with U being the genome wide mutation rate. Even if only one deleterious mutation enters a population each generation, the mutation free equivalent would have $\sim 63\%$ higher fitness. Now, this approximation considers only the mutation rate regardless of fitness effect and ecological context like density dependence (Mylius & Diekmann, 1995). Since then, a large body of literature has emerged, investigating the different mechanisms that affect the introduction and establishment of (deleterious) alleles. In the following, three of the most well established and supported key ideas will be discussed: the influence of allelic dominance in combination with the chromosomal location of an allele and the sex it is expressed in. For simplicity's sake, females will be referred to as the homozygous sex (having XX sex chromosomes) and males as the heterozygous sex (having XY sex chromosomes). The concepts however apply to systems in which the zygosity is switched, e.g., in organisms with ZZ males and ZW females. Charlesworth et al. considered that when selection acts on both sexes and dosage compensation exists, the relative rate of evolution of a beneficial X-linked allele, i.e., the rate at which a new beneficial X-linked mutation (allele) fixes in a population compared to an autosomal allele of equivalent effect, can be described as

$$R_X \approx 4h(2h + 1) \text{ (Equation 2a)}$$

with h being the level of dominance of the allele (Charlesworth et al., 1987). When $h < 0.5$ ($h = 0$ being a completely recessive and $h = 1$ being a completely dominant allele), the allele will fix relatively faster on the X chromosome compared to if it was located on an autosome. This has been labelled the faster-X theory. Second, they proposed that when selection acts on both sexes (Equation 8a) or only males (Equation 9) and dosage compensation exist, the relative rate of evolution of a slightly deleterious X-linked allele can be described as

$$R_X \approx 1 + \frac{1}{3}N s_1 \left(h - \frac{1}{2} \right) \text{ (Equation 8a),}$$

$$R_X \approx R_Y \approx 1 + \frac{1}{3}N s_3 \left(h - \frac{1}{2} \right) \text{ (Equation 9),}$$

with N being the effective population size, s_1 the fitness effect in females, and s_3 the X-linked fitness effect in males (in both cases $|N s_i| \ll 1$ is assumed). Reversely, this showed that when $h < 0.5$ the fixation of a deleterious X-linked allele is hindered. These equations are based on previous work such as from Haldane or Kimura but are the product of two central ideas. For one, that is that the X chromosome spends two thirds of the time in females but only one third of the time in males. Therefore, in theory, female-beneficial alleles should be favoured by selection regardless of dominance. Opposite, is the second aspect, which is that if an X-linked allele (located outside of the pseudo-autosomal region) is (partially) recessive it ‘behaves’ like an autosomal allele in females but will always be exposed to selection in males, regardless of dominance. Thus, female-deleterious alleles may stay hidden and could therefore accumulate on the X chromosome but should be less likely fixed if male-deleterious. As mentioned before, these findings are based on selection acting on both sexes or only males or females. In species with separated sexes, differences in males and females are widespread and diverse (Cox & Calsbeek, 2009). Sexual dimorphisms do not reflect concordance between the sexes, but conflict caused by antagonistic behaviour (interlocus sexual conflict) and opposing selection pressures (intralocus sexual conflict) (Bonduriansky & Chenoweth, 2009; Schenkel et al., 2018). The optimal phenotype in one sex may not be the same as it is in the other and is a product of past selection that has operated differently on each sex. For traits genetically correlated between the sexes, such that loci segregate for alleles that affect the fitness of both sexes, adaptation towards each peak may cease to be possible. Previous to work of Charlesworth et al. discussed earlier, Rice examined the conditions that a new and rare X-linked allele must meet in order to invade a population (Rice, 1984). He suggested that

$$S > \frac{2hT}{1 - hT} \text{ [Equation 3],}$$

with S being the fitness gain in the heterozygous sex and T the loss of fitness in the homozygous sex. If the allele is recessive enough, the heterozygous advantage quickly outweighs the homozygous disadvantage and an increase in frequency of the antagonistic allele becomes possible. The logic here too, is that a (partially) recessive allele located on the X chromosome may stay hidden from selection when expressed in females but is always exposed when expressed in males. In summary these concepts aid the idea that every individual accumulates some number of deleterious mutations, either through random drift or sex-specific selection pressures, and that these numbers may vary depending on whether the chromosome at question is an autosome or sex chromosome. The total loss in fitness caused by these deleterious variants has been labelled the mutational or genetic load, which consists of the realised loss in fitness caused by expressed deleterious variants and the potential loss in fitness caused by masked deleterious variants (Bertorelle et al., 2022). Whereas the loss in fitness caused by expressed deleterious variants seems obvious, the loss in fitness caused by masked variants is the potential threat they pose if located on a homozygous locus. The masked load may quickly become realised in the case of small population size and (or) bottleneck effects leading to consecutive inbreeding and increased homozygosity (Keller & Waller, 2002; Smeds & Ellegren, 2023).

This concept of hidden and exposed allele effects can be transferred to the eusocial and haplodiploid insects of the hymenopteran order. In insects like ants, the sexes are separated into different castes, each with a unique morphology and specific purpose inside the colony (Villet, 1992). Speaking very generally, a colony houses one queen which produces offspring, sterile female workers which are responsible for food acquisition, brood care and protection of the nest among other things, and males which provide gametes to other queens. Importantly, females are diploid whereas males are haploid. For the purpose of this study, males can be seen as an equivalent to the X chromosome (Kraaijeveld, 2009). Whereas, in a diploid system the male is hemizygous for the X chromosome, in a haplodiploid system the male is hemizygous for every chromosome. According to the masking theory, both X chromosomes and male hymenopterans should be subject to stronger purifying selection, as the corresponding genes are found in a haploid state (Cervantes et al., 2023; Kondrashov & Crow, 1991). Consequently, differences in substitution rate are expected when comparing the haploid to the diploid state (N. D. Singh et al., 2008).

Here we investigate the outcome of purifying selection in *Drosophila melanogaster* by comparing the genotypes found on the X chromosome to those found on the autosomes. We use the X chromosome of *D. melanogaster* as a proxy for the male hymenopteran, as it allows us to create even conditions between the genes expressed in a haploid condition and those in diploid

condition. By using genomic data from highly inbred *D. melanogaster* lines, we calculate the genetic load caused by putatively deleterious variants in a state where the majority of masked variants has become exposed. We use two independent methods to identify putatively deleterious variants, based on functional annotation of variants and constrained substitutions across multiple fly species phylogeny.

Methods

Genotype pre-processing

To study the effect of deleterious mutations on fitness, we investigated *Drosophila melanogaster* genotypes obtained from the Drosophila Genetic Reference Panel (DGRP) (Mackay et al., 2012) (formerly hosted on <http://dgrp.gnets.ncsu.edu/>). The DGRP provides genomic variant data on 205 *Drosophila* lines representing 20 generations of sibling matings to raise homozygous flies via inbreeding. We filtered the genotypes with PLINK (Version 1.9 beta) (Purcell et al., 2007) to remove variants that were represented in less than 10% of all lines. Subsequently, we used Beagle (Version 5.4) (Browning et al., 2018) to impute missing genotypes and increase the number of available variants. To analyse how chromosome location affects the accumulation of deleterious variants we used a functional effect based and a phylogeny based approach.

Identification of putatively deleterious variants

For the first method, we identified variants with putatively deleterious effects by their estimated impact on the organism, following the approach of previous studies (Chen, Soulsbury, Hench, et al., 2025; A. Singh et al., 2023; Wong & Holman, 2023). We merged the genotypes with functional annotations, which have been compiled with SnpEff (Cingolani et al., 2012), and kept only the variants categorised as having high impact.

Complementary to identifying putatively deleterious variants based on their estimated biological impact, we investigated the evolutionary conservation across 123 insect genomes (mostly *Brachycera* and *Nematocera* species) referenced to *D. melanogaster*. We measured the Genomic Evolutionary Rate Profiling (GERP) by comparing the observed substitutions to the substitutions which would be expected under neutral selection across the 124 insect species. These calculations used the HKY85 substitution model (Hasegawa et al., 1985) with a transition/

transversion ratio of 2.0 and nucleotide rates calculated from the multiple alignment. Using this method we assume that sites with less observed substitutions may be more constrained as they cause more harm if mutated. We obtained multiple alignment files (MAF) per chromosome for the 124 insect species from the UCSC genome browser and the corresponding phylogenetic tree files in Newick format (<https://hgdownload.soe.ucsc.edu/goldenPath/dm6/multiz124way/>). We converted these MAF files into FASTA files using PHAST (Version 1.9.7) (Hubisz et al., 2011) (Capella-Gutiérrez et al., 2009). Using the FASTA files and the phylogenetic tree as a reference we run the gerpcol command of GERP++ (Version 2.1) (Cooper et al., 2005; Davydov et al., 2010) (formerly hosted on <http://mendel.stanford.edu/SidowLab/downloads/gerp/index.html>) to calculate the neutral rate and rejected substitutions per site. Putatively deleterious variants, that is constrained variants, were classified by a rejected substitution score of ≥ 4 . Due to computational limits, we were not able to evaluate the GERP scores with the gerpelem command, which would have determined elements with significantly more rejected substitutions than what would be expected under neutral selection.

Both the SnpEff and GERP data set was then transferred to RStudio (Version 4.3.1) where the remainder of the analysis took place. Both data sets were split into a X chromosome set and one autosome set (2L, 2R, 3L, 3R, 4 chromosomes).

Calculation of genetic load

For both the SnpEff and the GERP data sets we calculated the homozygous, heterozygous and total population load per line for each chromosome type (X and autosome), following the methods of Chen et al. (Chen, Soulsbury, Hensch, et al., 2025):

$$\text{Total load}[i] = \frac{L_{HM} + 0.5L_{HT}}{L_T}$$

$$\text{Homozygous load}[i] = \frac{L_{HM}}{L_T}$$

$$\text{Heterozygous load}[i] = \frac{L_{HT}}{L_T}$$

The genetic load per *Drosophila* line i , i.e., the total load per line was calculated as the sum of the homozygous and heterozygous loci (with the latter being weighted by half), divided by the total number of loci. The homozygous load per line was calculated by summing up the homozygous loci and then dividing by the total number of loci, whereas the heterozygous load per line used the heterozygous loci instead. Because the data originated from highly inbred, and therefore, almost completely homozygous individuals for each loci, the genetic load mirrored the homozygous load closely. To take differences in chromosome group size into account, we standardised each load value by calculating the corresponding z-score, as follows:

$$Z\text{-score}_{[i,j]} = \frac{\text{Mutation load}_{[i,j]} - \mu}{\sigma}$$

The genetic load was Z-standardised for each genotyped locus i and for both types of genetic load j (SnpEff and GERP) using the mean across all lines μ and the standard deviation σ . To evaluate the impact of mutation load on fitness we downloaded fitness essays produced by Wong & Holman, who have measured reproductive success and survival in ‘early-life’ (2 to 5 days) and ‘late-life’ (14 to 17 days) males and females across 125 of the DGRP lines (Wong & Holman, 2023).

Modelling the relationship between fitness and genetic load

To assess the impact of genetic load on fitness alongside the influence of sex, chromosome type, we build multi-variate models using the brms package (Version 2.23.0) (Bürkner, 2017). We modelled fitness as a function of mutation load interacting with sex and chromosome type and added the line of origin as a random effect (Figure 1). To investigate pleiotropic effects between early- and late-life mutation accumulation, we build one model for each life stage. Fitness and mutation load were provided as continuous variables, sex and chromosome type as two-leveled categorical variables and line as a 125-leveled categorical variable. We assumed a normal distribution and chose a weakly informative prior to limit unreasonable results. To check whether the models fitted the data better when using unstandardised or standardised genetic load, we compared the model types for young and late stage to each other via the Leave-One-Out (LOO) cross validation function of the brms package. In all cases the validation reports suggested higher predictive power when using unstandardised genetic load as an explanatory variable (Supplementary, Table 1). We report the regression between mutation load and fitness, the posterior distribution estimations of our models, as well as differences in effect size between the sexes and life stages. For various methods of data manipulation we used the tidyverse package (Wickham et al., 2019) which includes ggplot for data visualisation, and patchwork to arrange plots (Pedersen, 2025).

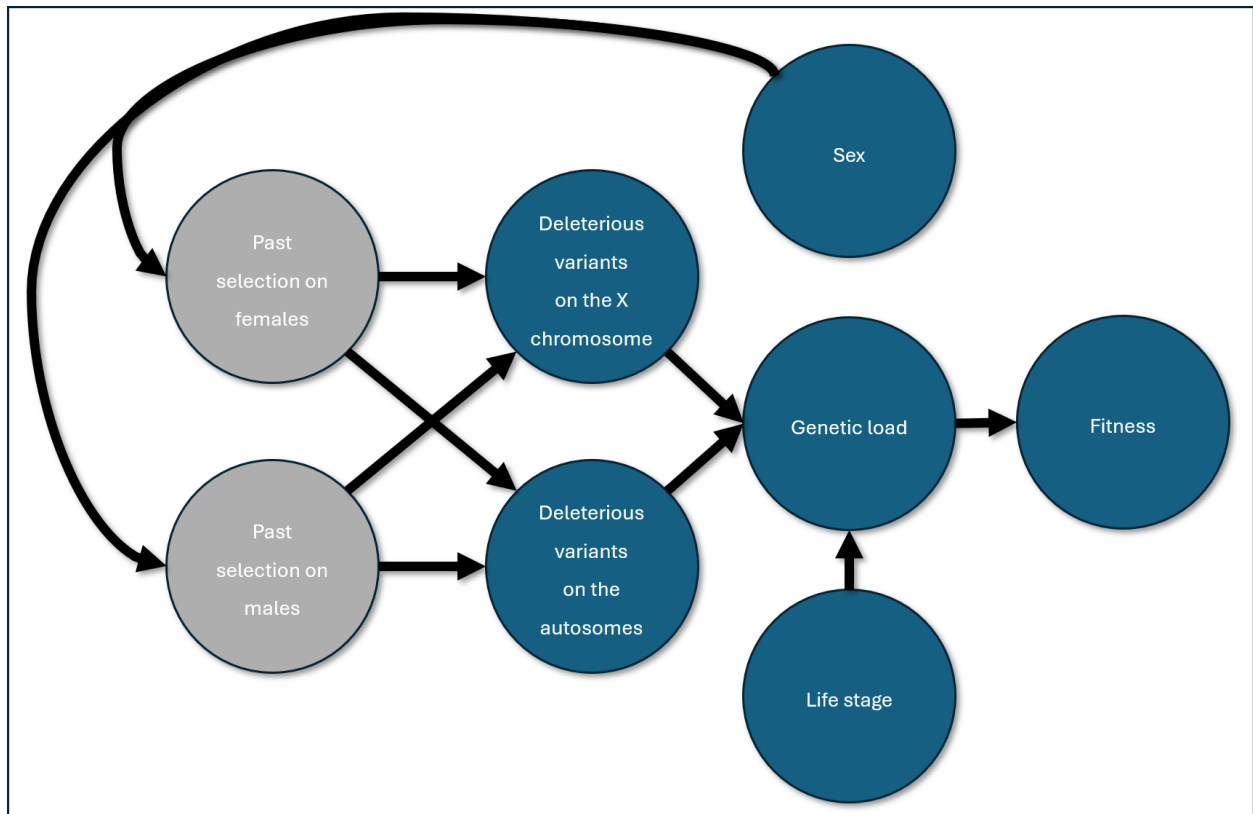


Figure 1: Visualisation of model structure and factors shaping fitness. Directed acyclic graph (DAG) of the influence of factors in relationship to fitness. Depending on the sex, selection acts differently on males and females, therefore, attributing to the accumulation of deleterious variants on the X chromosome and autosome. These deleterious variants in combination with the age and life stage of the organism cause genetic load which is expected to influence fitness. Blue points are represented as data in this project and used to approximate selection pressures, coloured grey.

Results

Influence of high impact deleterious variants on fitness

To examine if we find higher numbers of putatively deleterious variants in the autosomes than in the X chromosome, we obtained the genotypes of 205 inbred *Drosophila melanogaster* lines. After we imputed missing sites, we acquired a total of 4.026.614 variants across all chromosomes

except the Y chromosome. We filtered these variants by their estimated functional impact and found 282 high impact variants located on the X chromosome and 3.885 on the autosomes (Supplementary, Figure 1). We assume that the alternative alleles of the high impact variants are putatively deleterious, as the mutation of a high impact allele is likely disadvantageous.

As consecutive inbreeding is expected to affect heterozygosity, we investigated the number of heterozygous genotypes with high impact and for those with low or moderate impact. We found 11 heterozygous sites with high biological impact and 4.871 heterozygous sites with lesser impact (Supplementary, Figure 2). These heterozygous sites made up less than 0.2 % of all sites. For each *Drosophila* line we calculated the homozygous and heterozygous genetic load of the putatively deleterious variants by summing up the genotypes homozygous for the alternative allele, respectively, summing up the heterozygous genotypes, and dividing them by the total number of loci. We report the product of these two loads as the genetic load. As the heterozygous genotypes only contained one alternative, putatively deleterious allele, they were weighted by half. The genetic load was higher on the X chromosome compared to the autosomes, although overall low for both (Supplementary, Figure 2 A, B). When genetic loads were standardised to account for the lower number of deleterious alleles found on the X chromosome, both the X chromosome and autosomes showed similar levels of genetic load.

We used bayesian statistics to model the relationship between genetic load, the chromosome type, sex and their impact on fitness (Figure 2). We found little evidence that higher genetic load causes a reduction in fitness for both male and female *D. melanogaster* flies, both during their early and late life. Similarly, no differences in the impact of putatively deleterious variants was found between the X chromosome and autosomes. When comparing the model effect sizes between sexes and life-stages little variation was found.

Genetic load in *Drosophila melanogaster*

Impact of genetic load on fitness [SnPEff filter, non-standardised]

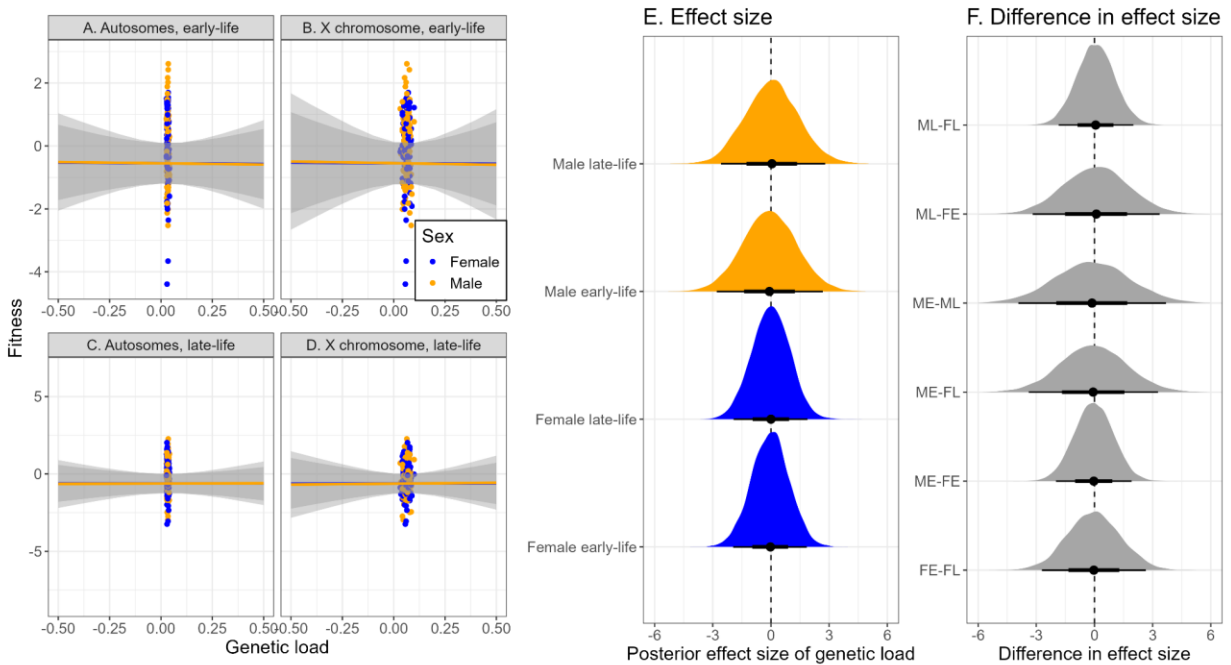


Figure 2: Impact of genetic load on fitness in the X chromosome and autosomes of early- and late-life *D. melanogaster*. The scatterplots (A-D) show the relationship between mean fitness and genetic load across 125 DGRP lines in males (orange) and females (blue) for both life-stages and chromosome types. The genetic load has been calculated for putatively deleterious variants, which have been identified by functional impact. Plot E shows the posteriors estimates of the regression slopes as well as the median and 66% and 95% quantiles in black. Plot (F) shows the differences in effect size between different combinations of sex and life stage. For the version of these plots using standardised genetic load see Supplementary, Figure 3.

Identification of constrained elements on the X chromosome and autosomes

Complementary to determining putatively deleterious variants via their estimated impact, we identified locations in the *D. melanogaster* genome which were more constrained than what would be expected under neutral selection. For this we compared the rate of rejected substitutions in the *D. melanogaster* genome to that rate of 124 other insect genomes (mostly Diptera species). Across a total of 4,075,700 sites, we found 196,374 highly constrained sites (rejected substitution score ≥ 4) on the X chromosome and 1,302,241 highly constrained sites across the autosomes. However, the general distribution of the rejected substitution scores was similar between the X chromosome and autosomes (Supplementary, Figure 4). We calculated the genetic load of the

constrained sites in the same way as described previously and ran bayesian statistics using the same structure. We did not see a reduction in fitness due to increasing genetic load of constrained elements (Figure 3). This was the case for both sexes, early and late life stages as well as the X chromosome and autosomes. Furthermore, we did not find strong differences between the different combinations of sex and life stage.

Impact of genetic load on fitness [GERP filter, non-standardised]

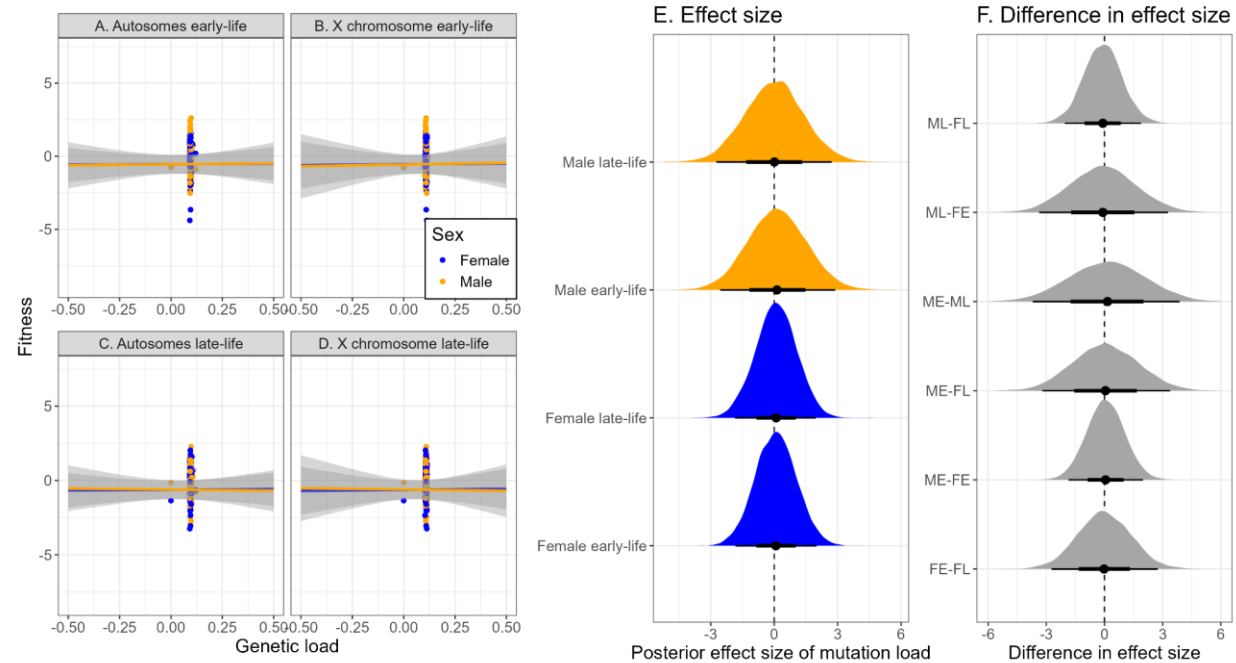


Figure 3: Impact of genetic load on fitness in the X chromosome and autosomes of early- and late-life *D. melanogaster*. The scatterplots (A-D) show the relationship between mean fitness and genetic load across 125 DGRP lines in males (orange) and females (blue) for both life-stages and chromosome types. The genetic load has been calculated for putatively deleterious variants, which have been identified by their rejected substitutions score compared to a neutral substitution rate. Plot E shows the posteriors estimates of the regression slopes as well as the median and 66% and 95% quantiles in black. Plot (F) shows the differences in effect size between different combinations of sex and life stage. For the version of these plots using standardised genetic load see Supplementary, Figure 5.

Discussion

In this study we investigated if the fitness of male and female *D. melanogaster* flies responds differently to increasing genetic load, a measurement of the accumulation of deleterious variants

in the genome. To analyse a state in which the effect of recessive deleterious alleles is exposed, we use genotypes from the DGRP, a project in which the genome of *D. melanogaster* flies has been made almost completely homozygous due to repeated sib matings (Mackay et al., 2012). We classified minor, alternative alleles as putatively deleterious based on two independent approaches. Firstly, based on the estimated functional impact a mutation of the locus causes, and secondly based on genomic sites showing increased numbers of rejected substitutions, suggesting higher selective constraint as a mutation of said genomic site may be disadvantageous.

Classical modelling studies have established the theory that the hemizygous sex, that is, males in *Drosophila* as they only carry on X sex chromosome, accumulate male-beneficial genes faster on the X chromosome as genotypes are always exposed to selection when found in males (Charlesworth et al., 1987; Patten, 2019). At the same time and for the same reasons the intrusion of male-deleterious genotypes is hindered on the X chromosome. Therefore, in a comparison one may expect to find an advantage in fitness in males in response to inbreeding depression compared to females, as females do not experience this ‘protecting’ effect from deleterious genotypes on the X chromosome.

We report low numbers of genetic load in general and find no evidence for such differences between males and females. In the following we discuss possible reasons for our results. Both theoretical and empirical studies have shown that opposite to the negative effects of inbreeding, i.e., especially the exposure of deleterious recessives and loss of genetic variability, positive effects may follow to a lesser degree. As deleterious recessive genotypes become unmasked they may be effectively removed from a population due to genetic purging (Crnokrak & Barrett, 2002; García-Dorado, 2012; Khan et al., 2021; Kleinman-Ruiz et al., 2022). This purging effect has been suggested to be even stronger when inbreeding depression is caused by lethal mutations (Hedrick, 1994) and extensive research has proposed that half of the inbreeding depression for viability in *Drosophila* is caused by recessive lethals (Hedrick & Garcia-Dorado, 2016; Simmons & Crow, 1977). In the case of the DGRP study, the 20 generations of sib matings may have been very effective in purging deleterious recessive genotypes, which could explain the low numbers of genetic load we were able to detect on both the X chromosome and autosomes. We found almost no heterozygous sites, which is problematic as genetic load is calculated on the occurrence of heterozygosity. However, previous studies have successfully used the DGRP data to calculate genetic load but report similar low numbers (Wong & Holman, 2023). Another caveat of using data of heavily inbred populations is that it is not possible to explore and compare homozygous, heterozygous and total genetic load to each other, as the

heterozygous load is basically non-existent and the total load, therefore, mirrors the homozygous load. This option has not necessarily been explored in studies using wild populations too though (Chen, Soulsbury, Hench, et al., 2025; Chen, Soulsbury, Oers, et al., 2025; Smeds & Ellegren, 2023). Another reason why we have found only low numbers of genetic load may be that genetic load was calculated only for those lines in which fitness has been measured. As the line mean fitness essays are based only on those flies that have survived both the early and late stages, those individuals with lethal or at least deleterious enough mutations to not survive until late life are not represented (Wong & Holman, 2023), making the estimation of deleterious genotypes even more conservative. The severity of the mutations caused by the successive inbreeding may also impede predictions, as theories like the faster-X effect assume slightly deleterious fitness effects of mutations.

Perhaps as a result of the circumstances above, we also find little difference in the response in fitness to ‘increasing’ genetic load between males and females and the X chromosome and autosomes. Besides low genetic load, it has been shown that the *Drosophila* X sex chromosome is strongly demasculinised, i.e., lacking male-beneficial genes (Gurbich & Bachtrog, 2008; Parisi et al., 2003; Sturgill et al., 2007). This is unlike the findings in most other organisms and opposite to what theory predicts. One aspect is that the X chromosome is located two thirds of the time in females and only one third of the time in males. This could benefit the accumulation of female-beneficial genotypes, especially if traits are dominant (Rice, 1984). Importantly, a genotype with sexually antagonistic effects could be realised into a phenotype via sex-biased gene expression which obscures the clear predictions based on theoretical modelling. A locus expected to be selected for may be down- or upregulated in one sex, changing how it is ‘perceived’ by selection. An example of this is the X chromosome inactivation during spermatogenesis, which has been suggested to make the X chromosome an inhospitable location for late but not early spermatogenesis genes (Wu & Xu, 2003). If this is actually the case in *Drosophila* has been debated though (Lu & Wu, 2011; Mikhaylova & Nurminsky, 2011).

Apart from biological reasons for why our findings may not match expectations, there are some technical caveats of this analysis that needs to be addressed. Per DGRP line, the fitness essays we used report mean line fitness per sex and life stage (Wong & Holman, 2023), i.e., four fitness entries per line. Although it was possible to match the location of variants on the chromosomes with functional annotations or rejected substitution scores, and therefore, calculate the genetic load for each line per X chromosome and autosomes, this information on chromosomes was not present in the fitness essays. Therefore, for each of the four combinations of sex and life stage in

one specific DGRP line, we report the same genetic load values calculated for putatively deleterious variants located on the X chromosome and autosomes.

Reversely, the fitness essays did use male and female flies to measure fitness, the genotypes that were available to us were based on females only. But this may be only to a lesser degree problematic, as information on the X-linked genotypes was still provided. A third technical constraint was that we could not explore the full potential of the genomic evolutionary rate profiling, which allows for the identification and clustering of elements with constraint above expectations. This was not due to data structure but computational limits. It would have been insightful to see if we found differences in the number and size of constrained regions between the X chromosome and autosomes.

Summary

Due to the hemizygous state of the X chromosome in males, it has been suggested as a favourable location for male-beneficial genotypes. An analogy between the X chromosome and the haploid males of eusocial insects has been drawn previously, proposing that these males should be subject to stronger purifying selection as well. We tested this hypothesis using genotypes and fitness measurements of heavily inbred *Drosophila melanogaster* flies, causing recessive deleterious variants to be exposed as a proxy. We found no relevant relationship between the accumulation of putatively deleterious variants and fitness, and no visible differences in said relationship between males and females and the X chromosome and autosomes. This may be due to biological reasons as the *Drosophila* X chromosome houses untypically low numbers of male-beneficial genes (demasculinisation), which perhaps makes

D. melanogaster an unsuitable candidate to analyse for this type of question. Complementary, the repeated inbreeding of the flies may have caused a strong purging effect of deleterious variants, explaining the overall low numbers of genetic load we report. However, our results can also be traced back to technical limitations which we discuss as well. Nonetheless, we provide a comprehensive workflow to reproduce these methods, which can be an important tool for future research. It is a fair assumption that this analysis yields different results if applied to an organism which lacks the traits peculiar to the *Drosophila* X chromosome, or data which allows for a better matching between genotypes and fitness measurements.

Supplementary

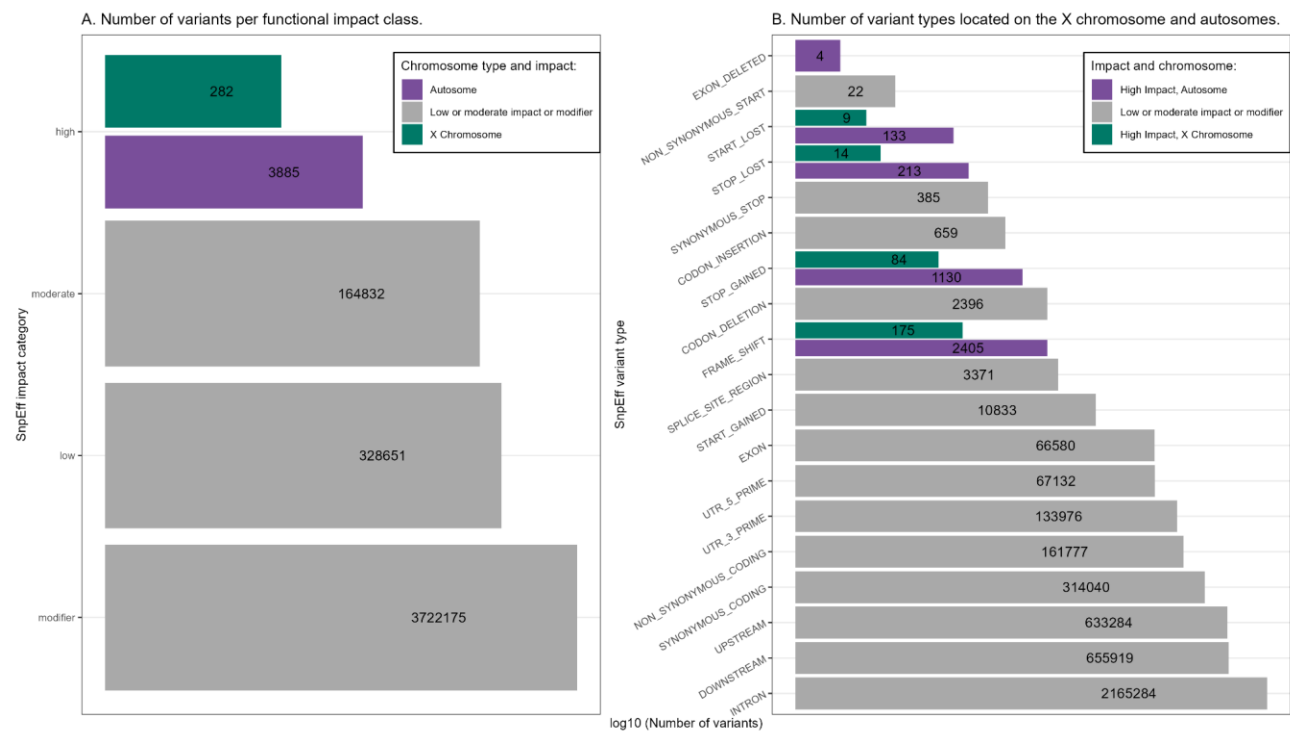
Detailed information on the methods can be found at <link>

Supplementary, Table 1: Summary of metrics of LOO cross validation between models using unstandardised and standardised genetic load values.

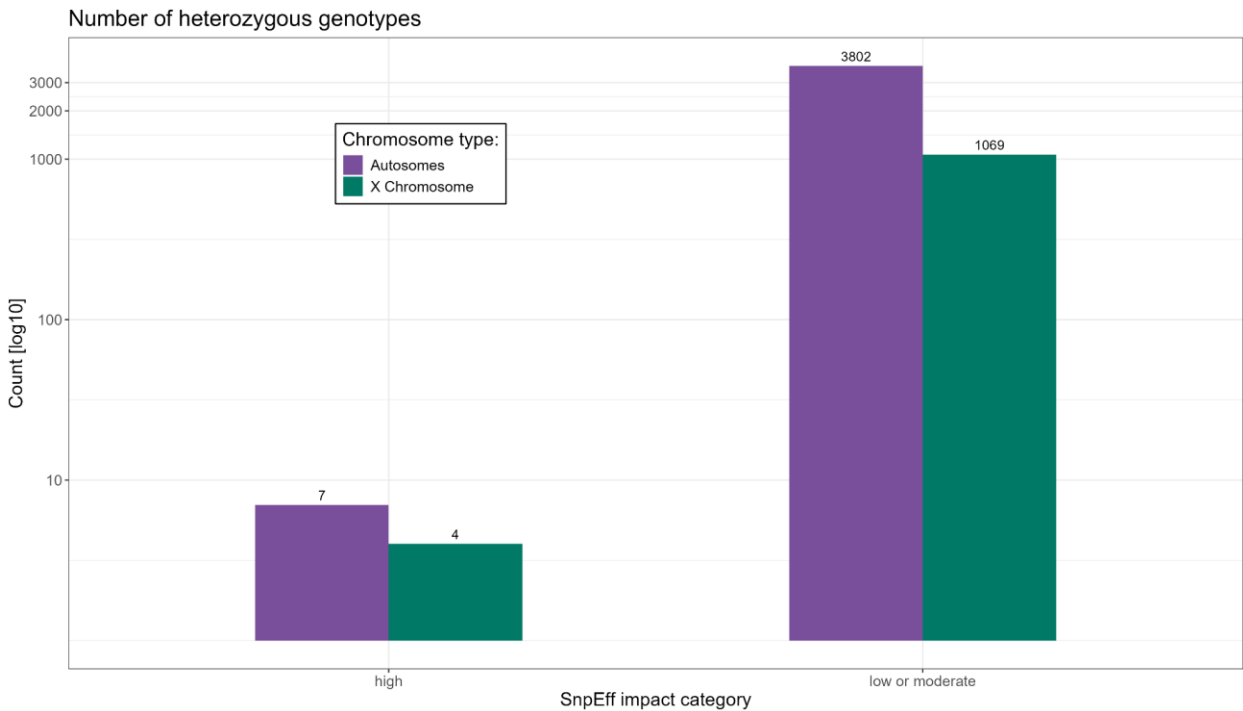
Life stage	Filter type	ELPD difference (unstandardised as reference)	SE difference (unstandardised as reference)	ELPD_diff / SE_diff
Early-life	SnpEff	-4.3	1.1	3.9
Late-life	SnpEff	-4.2	1.1	3.8
Early-life	GERP	-3.8	2.1	1.8
Late-life	GERP	-2.3	1.3	1.8

The table shows comparison metrics between models using unstandardised and standardised genetic load. The third column shows the difference in ‘Expected Log Predictive Density’ between the models. In all cases, the models using standardised genetic load values had lower (negative) ELPD than when using unstandardised genetic load, thus having lower predictive performance. Similarly, all models had higher (positive) standard errors when using standardised genetic load than the unstandardised versions. The absolute quotient of ELPD difference divided by SE difference approximates the performance between the models, with values ≤ 4 suggesting neglectable differences in predictiveness.

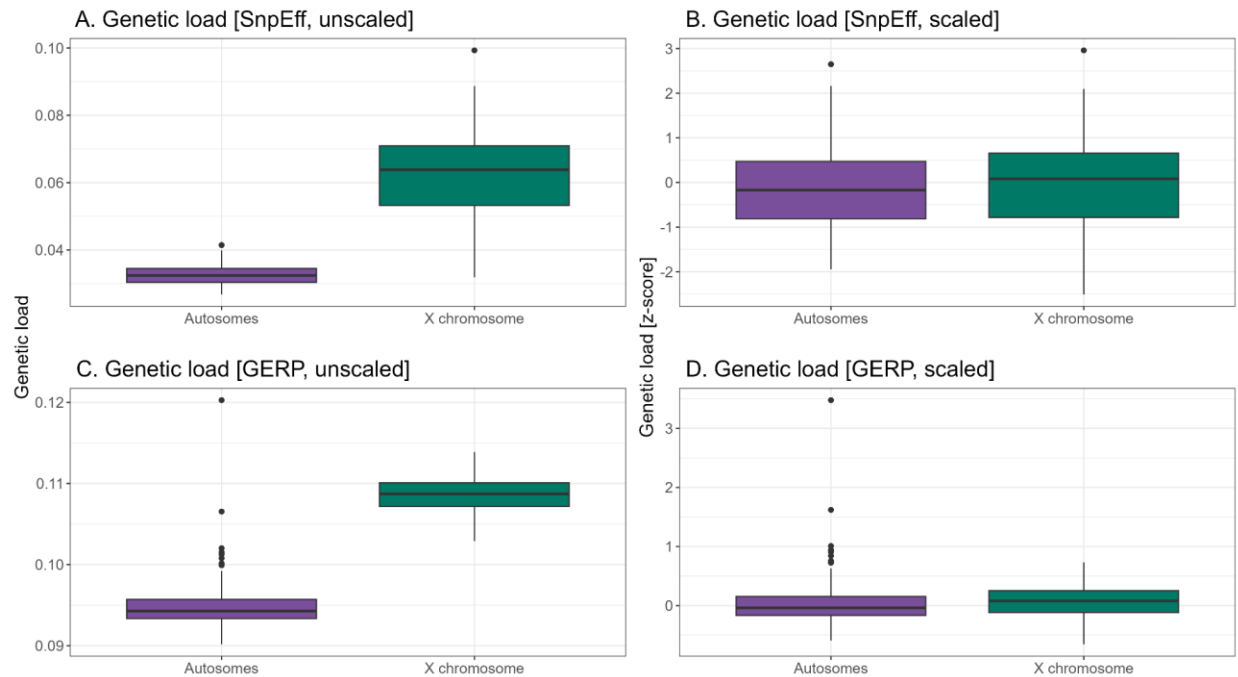
Genetic load in *Drosophila melanogaster*



Supplementary, Figure 1: Estimates of impact and function of putatively deleterious alleles. Bar plots of the estimated impact (A) and function (B) of variants annotated via SnpEff. The bars of variants estimated to have a high biological impact, thus putatively deleterious when mutated, as well as their respective functions are highlighted in green for the X chromosome and purple for the autosomes.



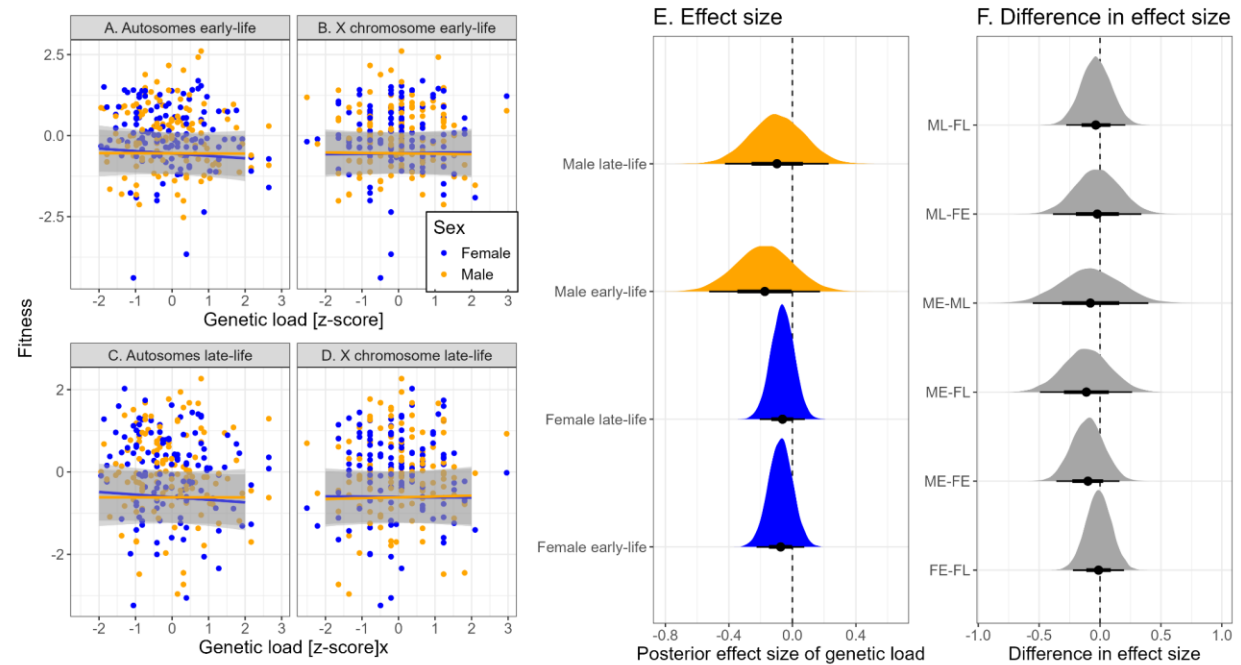
Supplementary, Figure 2: Numbers of heterozygous sites with high or lower impact. Bar plots showing the log10-transformed counts of heterozygous genotypes estimated as having high biological impact or at least moderate impact. The sites which are located on the X chromosome are coloured in green and those located on the autosomes in purple.



Supplementary, Figure 3: Genetic load of putatively deleterious alleles in the X chromosome and autosomes. Box plots of the genetic load caused by putatively deleterious alleles located on the autosomes (purple) and the X chromosome (green). To determine putatively deleterious alleles the methods of functional impact estimation via SnpEff (A,B) and calculation of constrained elements via GERP++ (C, D) was used. Shown are box plots for unstandardised (A, C) and standardised genetic load via z-score (B, D).

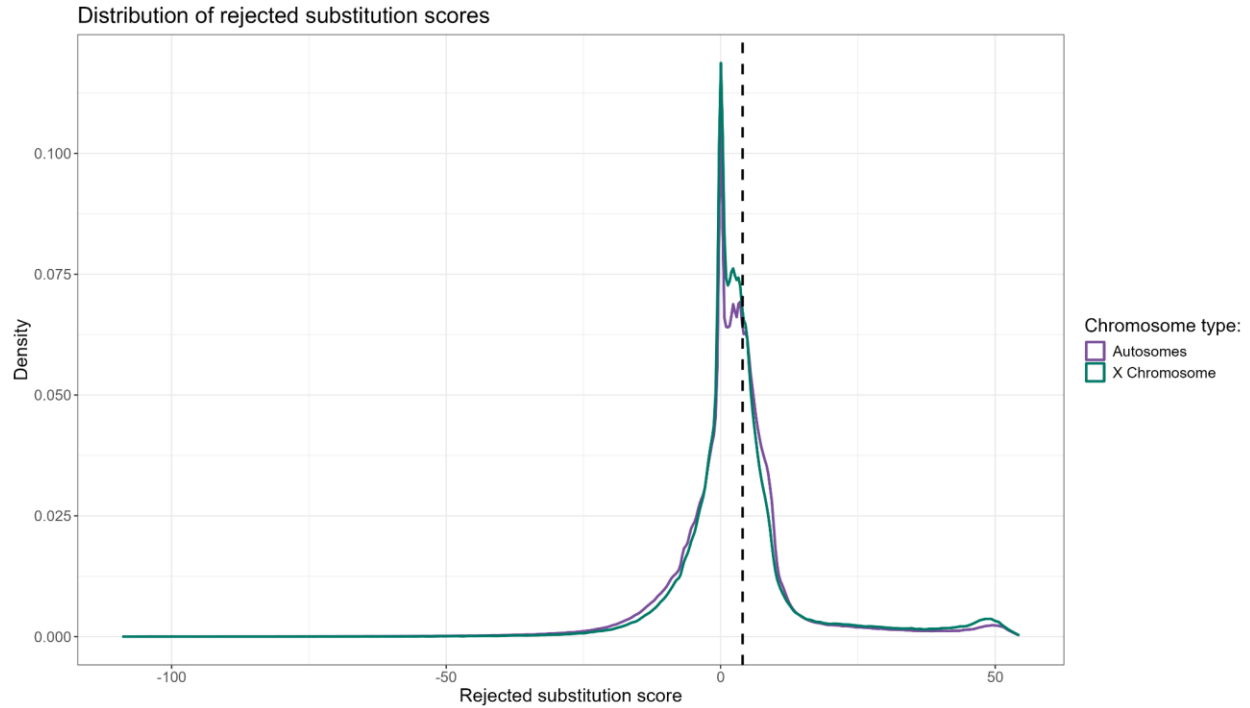
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Impact of genetic load on fitness [SnPEff filter, standardised]



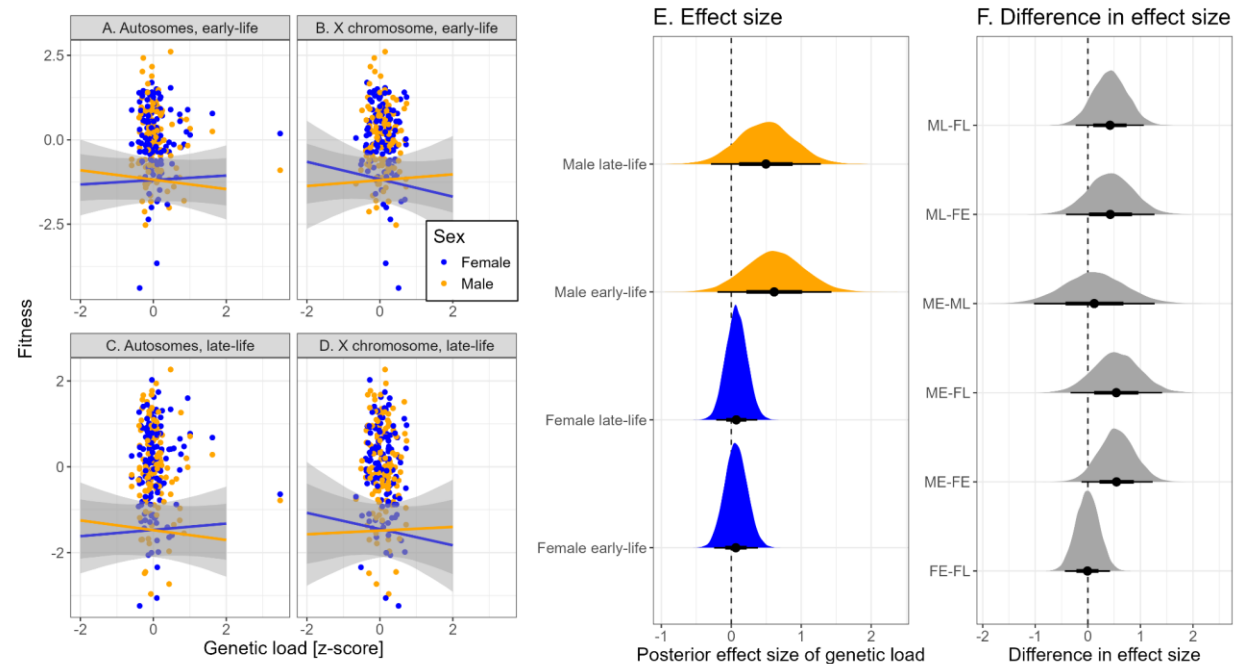
Supplementary, Figure 4: Impact of genetic load on fitness in the X chromosome and autosomes of early- and late-life *D. melanogaster*. The scatterplots (A-D) show the relationship between mean fitness and genetic load across 125 DGRP lines in males (orange) and females (blue) for both life-stages and chromosome types. The genetic load has been calculated for putatively deleterious variants, which have been identified by functional impact. The genetic load has been standardised via z-score (using the mean and standard deviation) to take difference in gene number between the X chromosome and autosomes into account. Plot E shows the posteriors estimates of the regression slopes as well as the median and 66% and 95% quantiles in black. Plot (F) shows the differences in effect size between different combinations of sex and life stage.

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Supplementary, Figure 5: Density distribution of rejected substitution scores. Density plot of the rejected substitution (RS) score calculated via GERP++ for the X chromosome (green) and autosomes (purple). The vertical, dotted line marks the RS scores ≥ 4 used to classify strongly constrained, and thus, putatively deleterious variants.

Impact of genetic load on fitness [GERP filter, standardised]



Supplementary, Figure 6: Impact of genetic load on fitness in the X chromosome and autosomes of early- and late-life *D. melanogaster*. The scatterplots (A-D) show the relationship between mean fitness and genetic load across 125 DGRP lines in males (orange) and females (blue) for both life-stages and chromosome types. The genetic load has been calculated for putatively deleterious variants, which have been identified by their rejected substitutions score compared to a neutral substitution rate. The genetic load has been standardised via z-score (using the mean and standard deviation) to take difference in gene number between the X chromosome and autosomes into account. Plot E shows the posteriors estimates of the regression slopes as well as the median and 66% and 95% quantiles in black. Plot (F) shows the differences in effect size between different combinations of sex and life stage.

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